# CASE STUDY 2 - Shared File System using AWS EFS

**Problem Statement**

ABC is India based entertainment production company with a focus on Northeast and East Indian cinema. They wanted a highly available and reliable storage solution for their onprem data. The client wanted the data to be reliable, highly available, secure, and persistent and also wanted to have data on the storage to be accessible from all the servers. The issues with the existing infrastructure are mentioned below:

1) They were using NAS storage which had limitations in scalability.

2) The client wanted a complete application to run in Aws cloud with storage in the cloud to keep his files in sync with servers so that he has centralized storage of data

3) There were security issues in the existing infrastructure and no encryption at rest or transit.

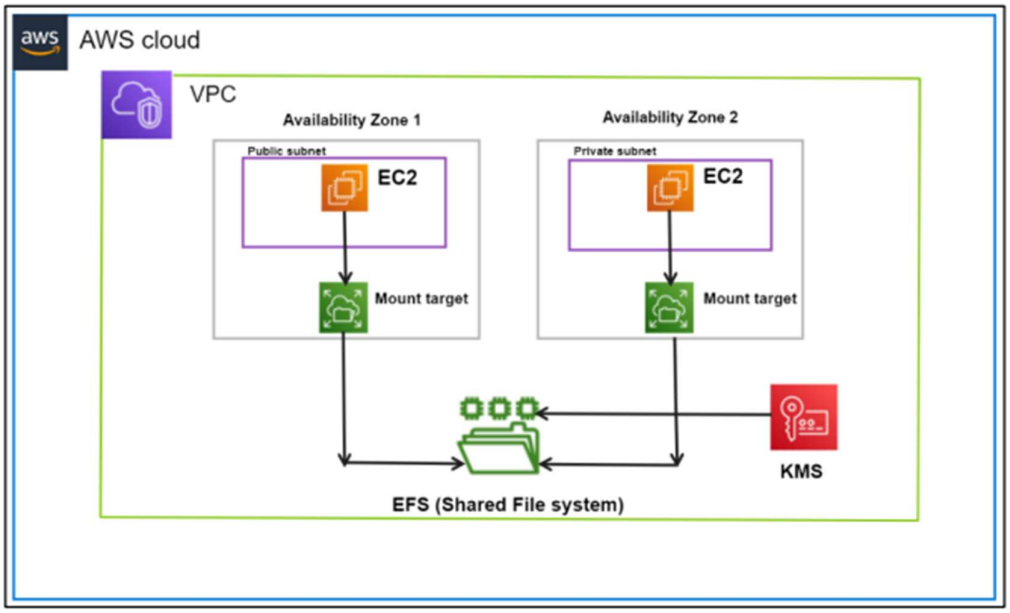
4) Manual intervention was needed to change the storage type and transfer files to infrequent storage.

5) The client is unable to scale up the infrastructure due to high capital costs for new hardware.

6) Need for low-cost storage options for both frequent and infrequent data.

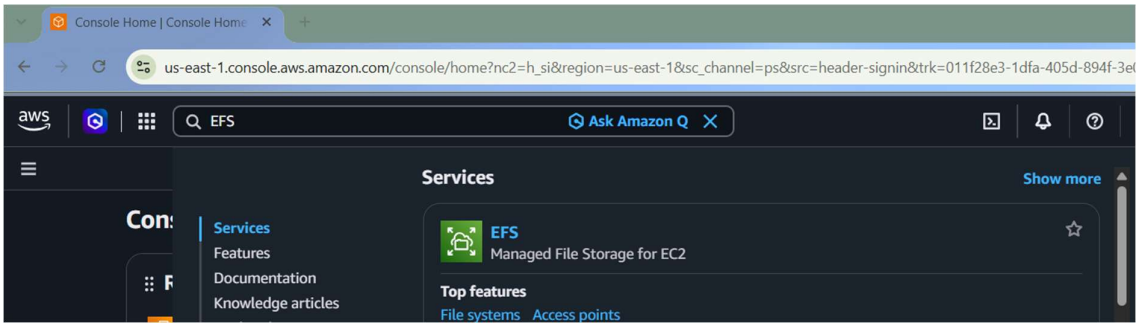
7) Highly available, secure, and persistent shared File system in AWS cloud with EFS.

## Architectural Diagram

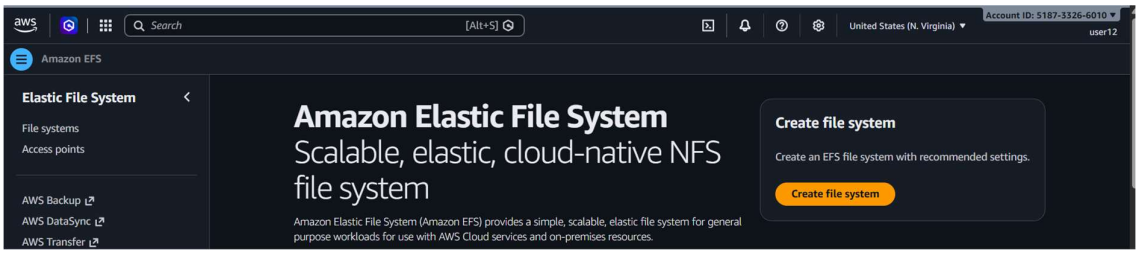


**Solution**

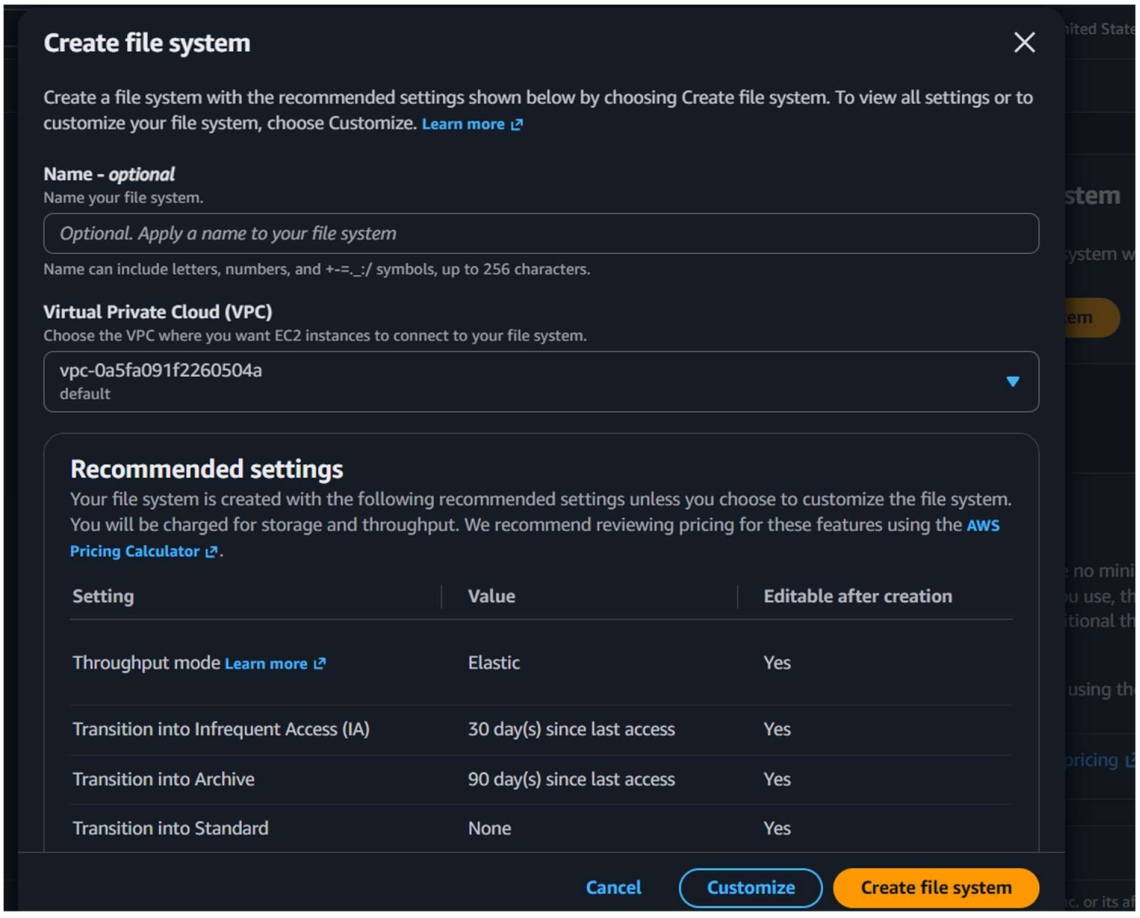
1. AWS management console- > select EFS from services.




2. EFS dashboard will open -> Click on Create file system.

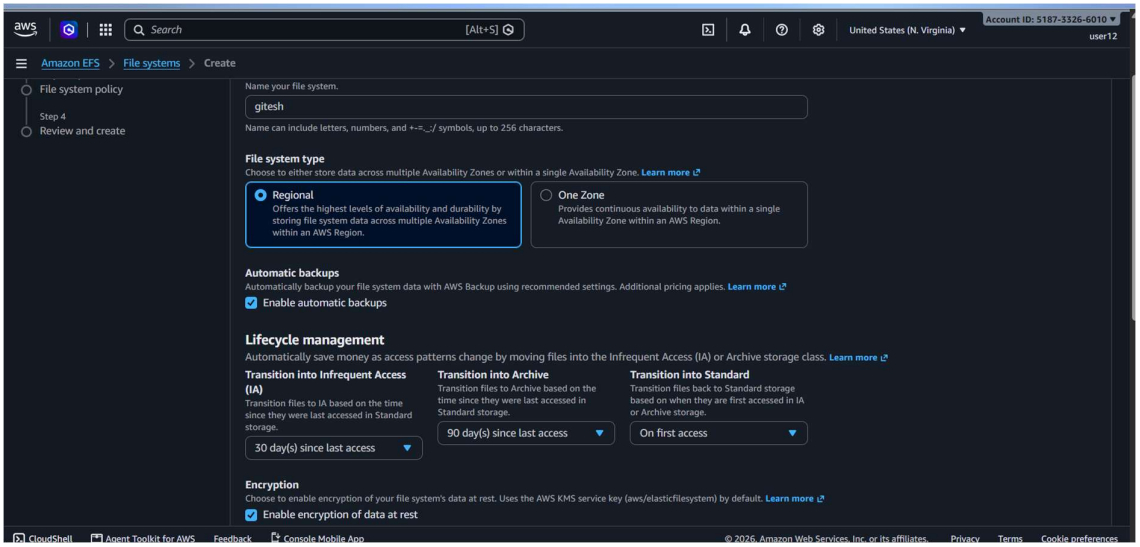



3. EFS create wizard will open -> Click on Customize

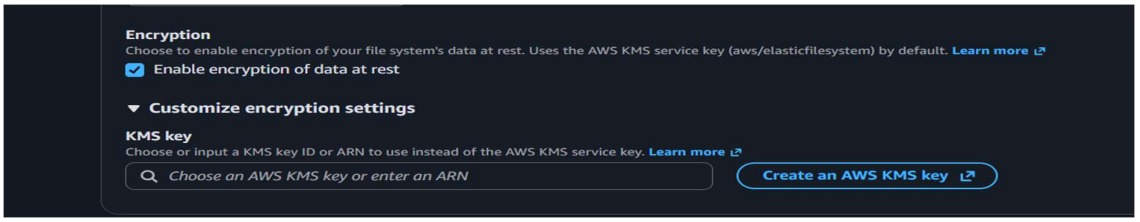


4. Enter the respective data as below:

a) Name = Name of your choice
b) Storage class = standard
c) Automatic backup = enabled
d) Transition to IA = Choose 30 days from the list
e) Transition out of IA = Choose On first access

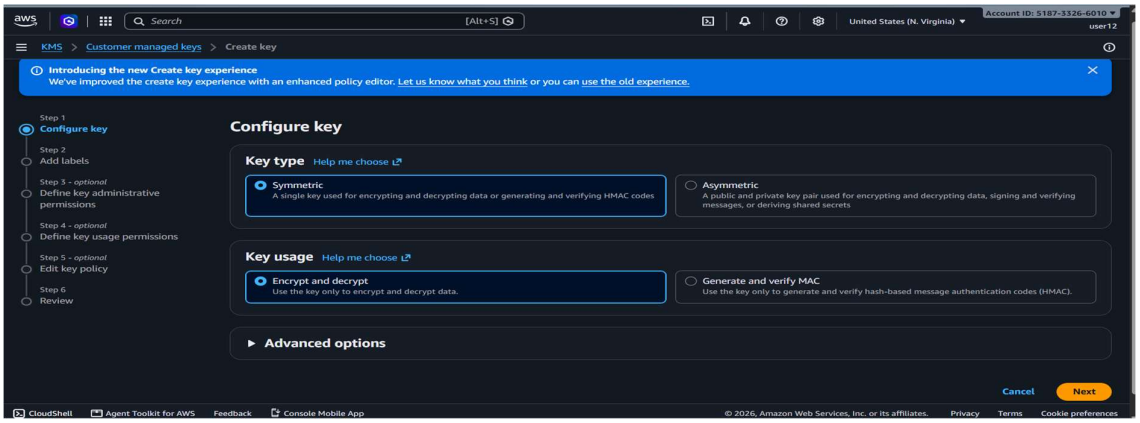



5. Enable encryption at rest, and in encryption settings, click on Create an AWS KMS key.



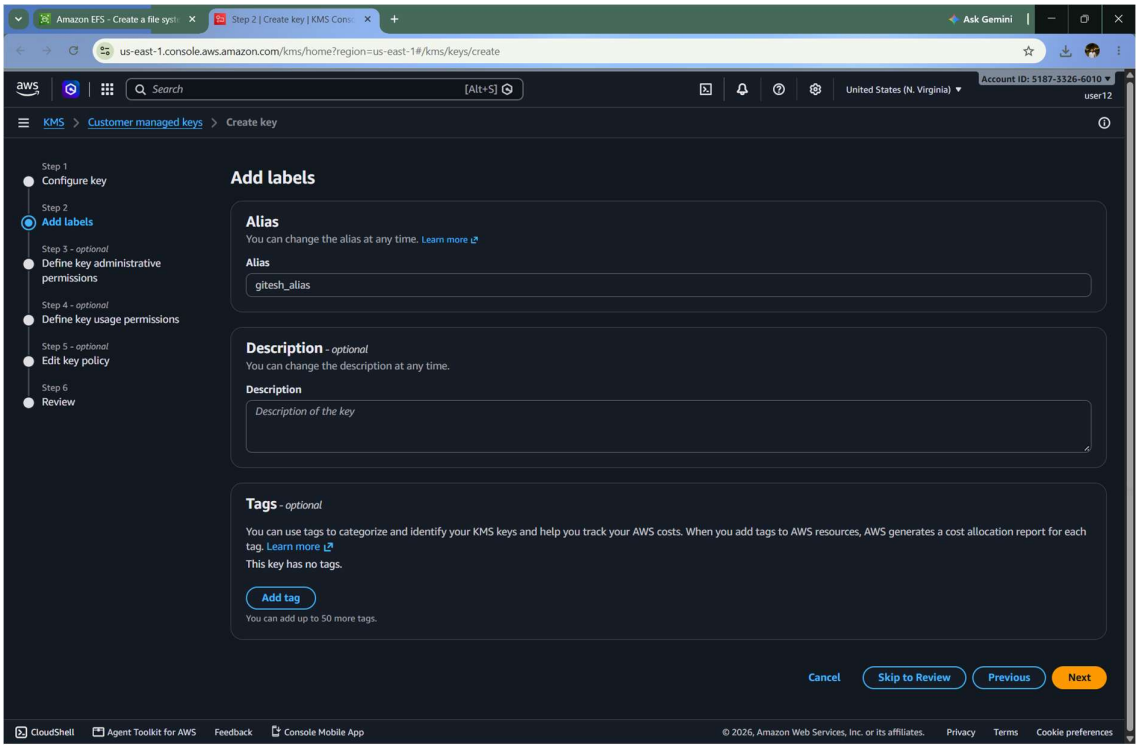

6. After the KMS dashboard is open. Enter the respective data as below:

a) Key type = Symmetric
b) Key usage = Encrypt and decrypt
c) Keep rest as default and click Next

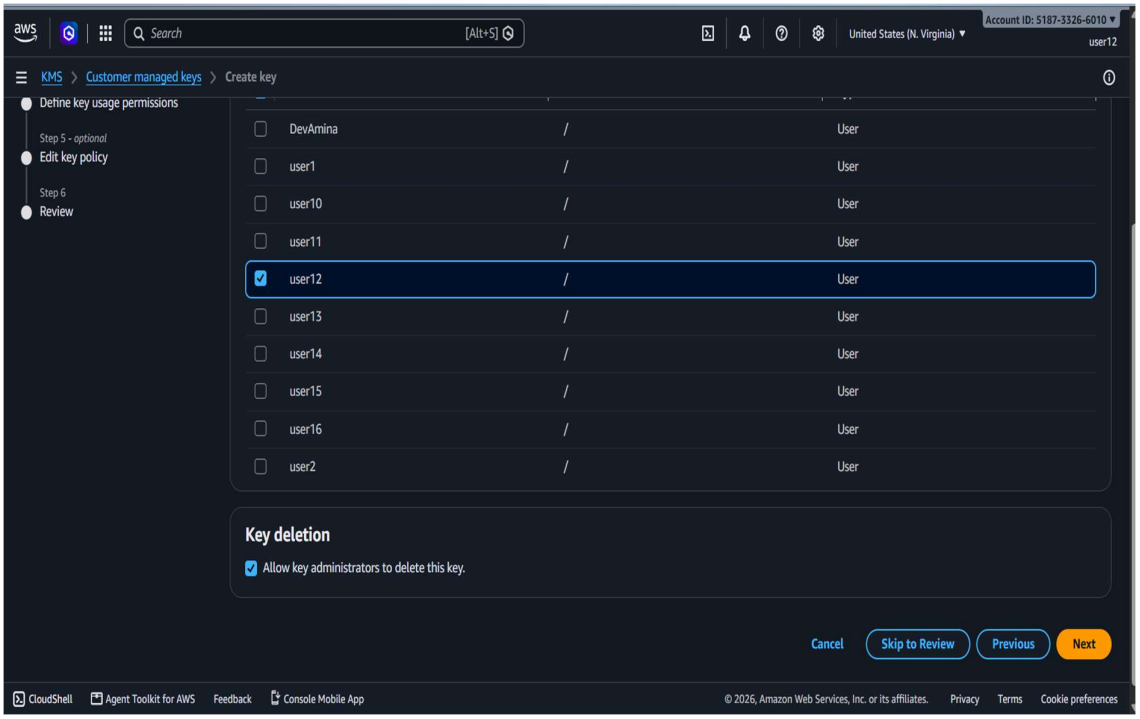


7. In Add labels, Enter Alias with a name of your choice.

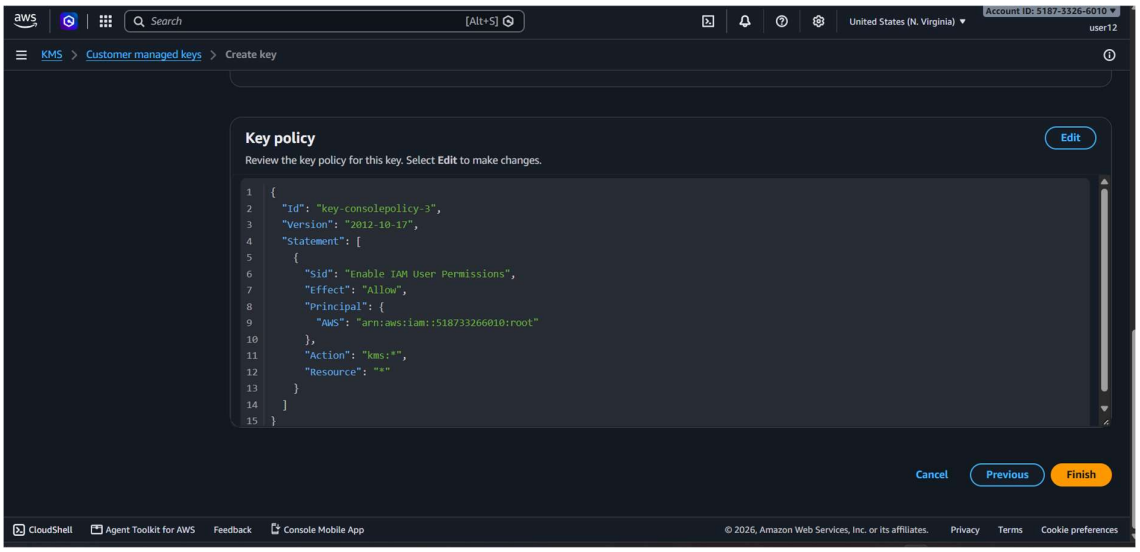
8. Keep rest as default and click Next.



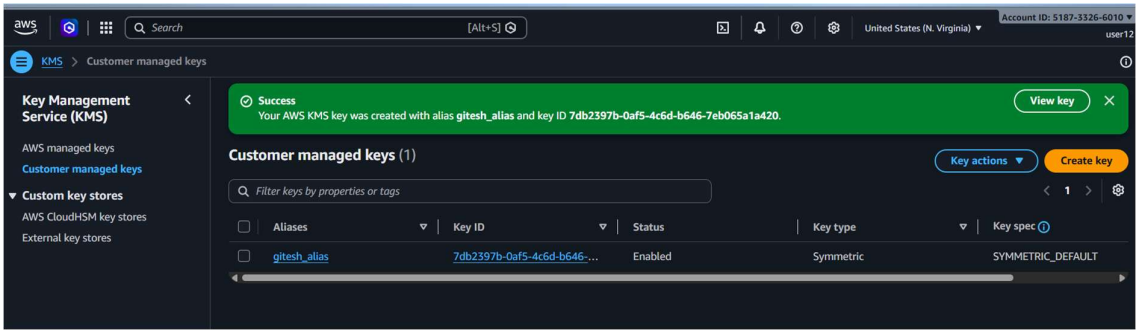

9. Leave key administrator and key deletion options as default and click Skip to review



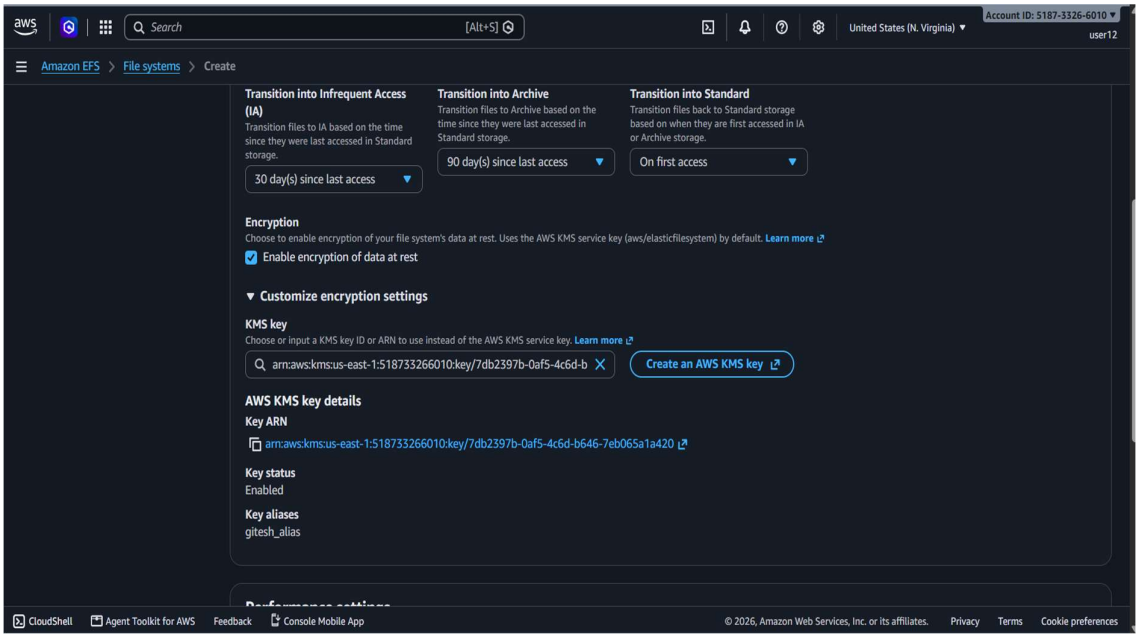
11. Review and Click on Finish



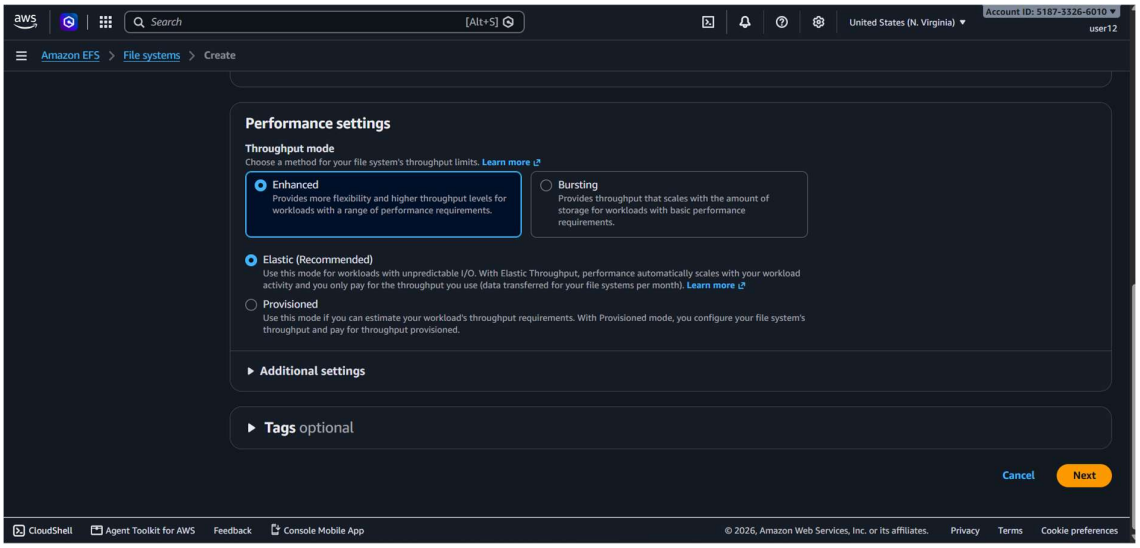

12. A Success prompt will be visible, and KMS Key is created.




13. Go back to EFS create wizard screen and Choose the KMS key created in previous step from drop down list.

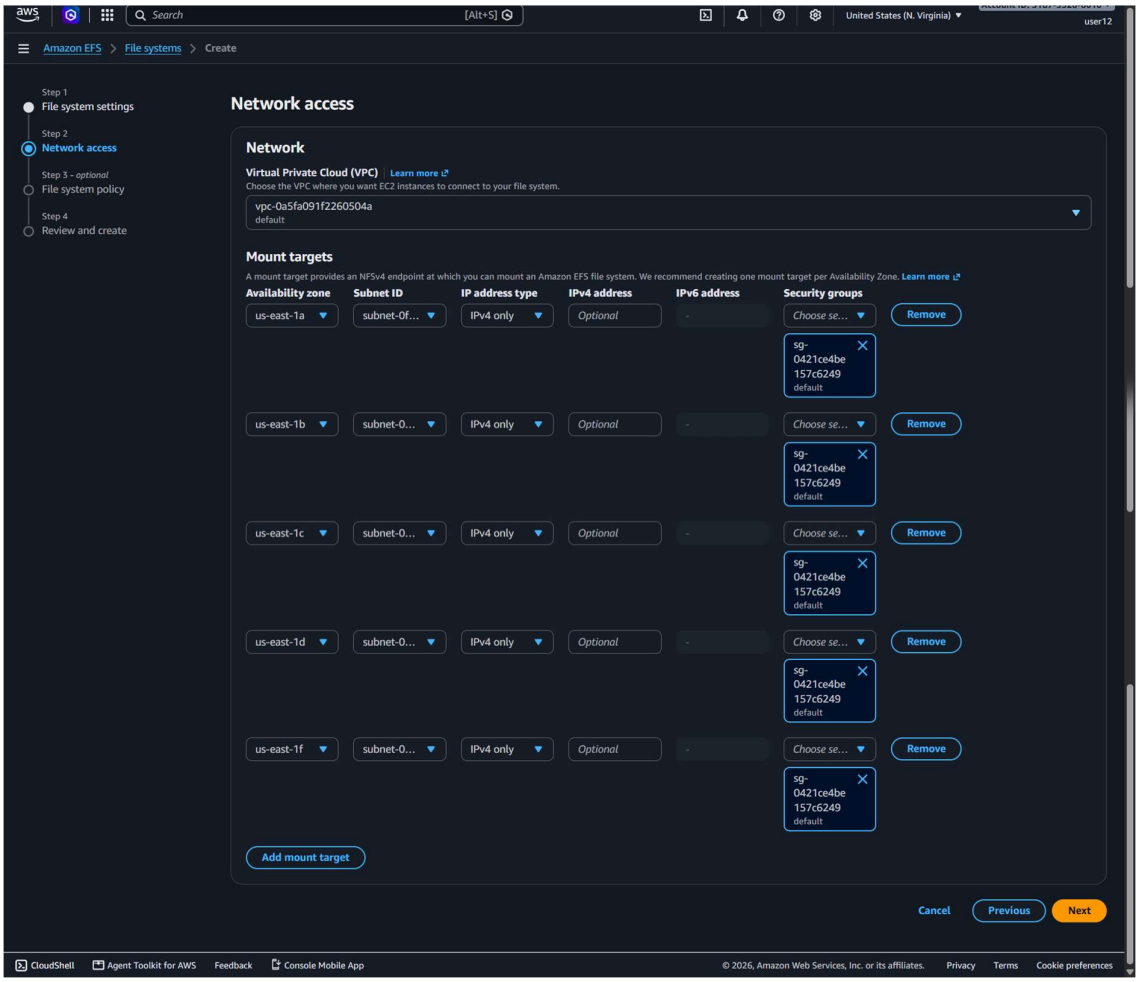


14. Leave rest as default and Click Next.

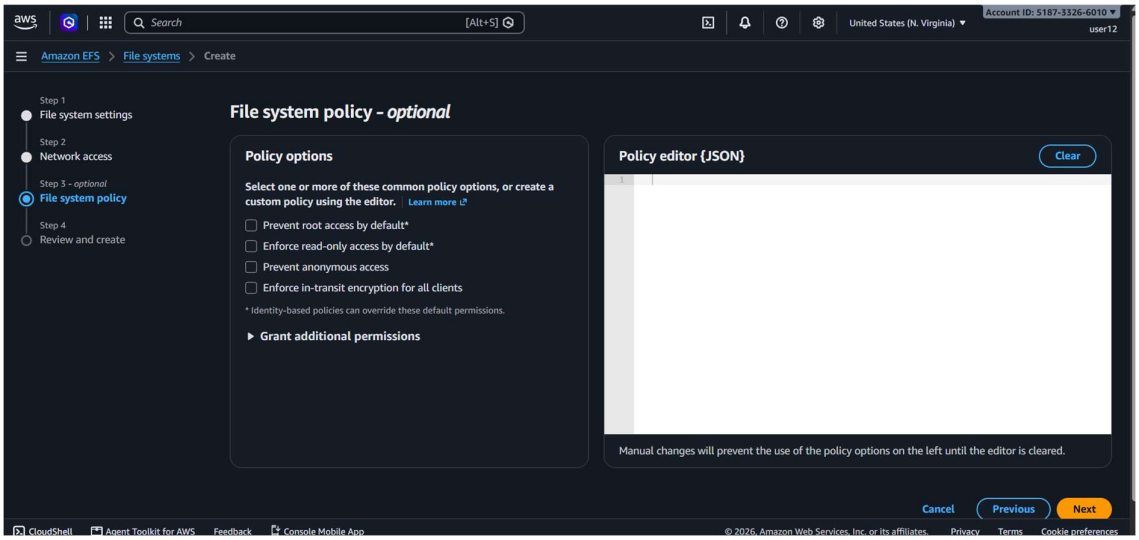



15. In Network access setting. Under VPC, Select VPC of your choice

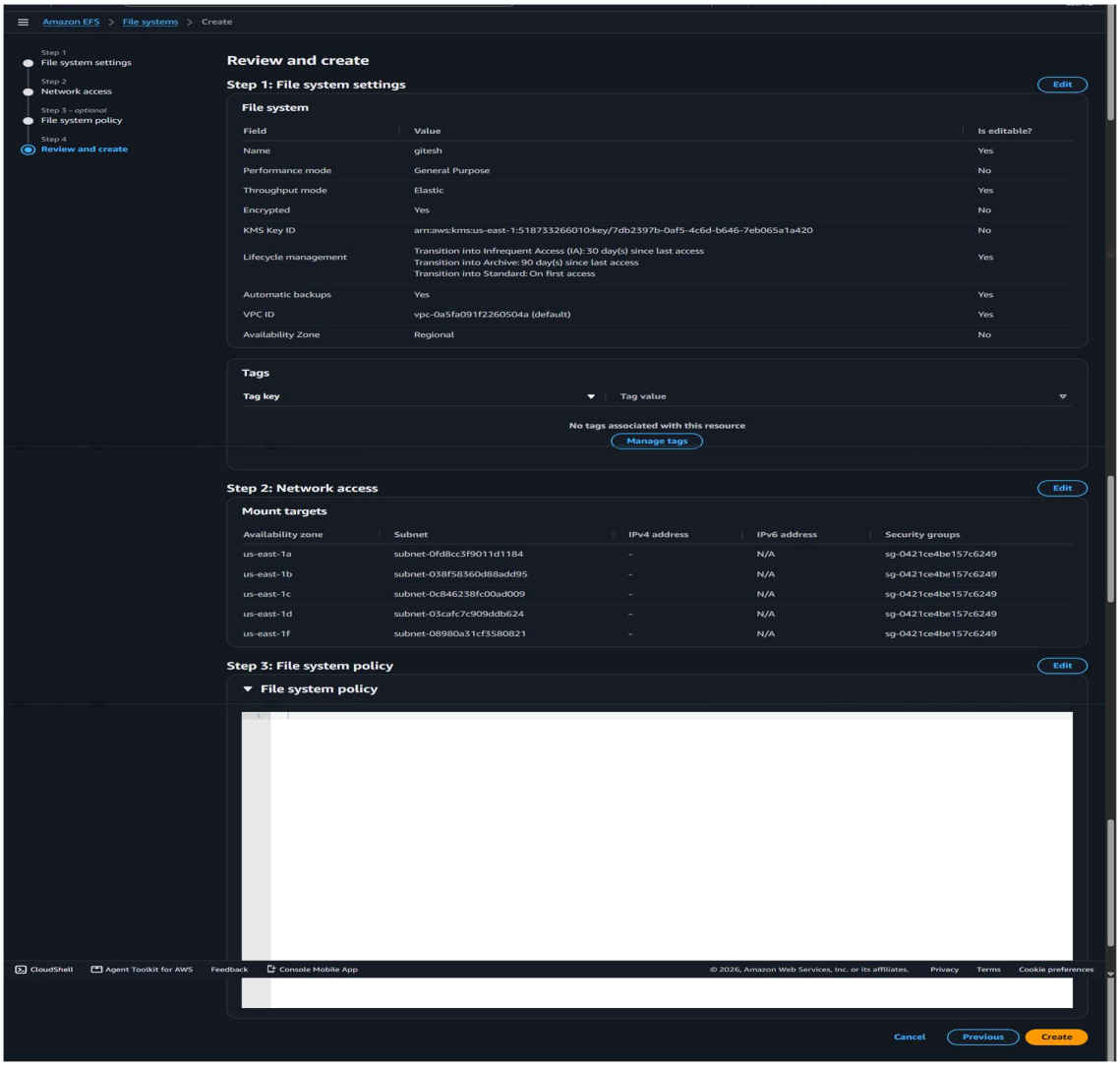
16. Leave rest as default and click Next



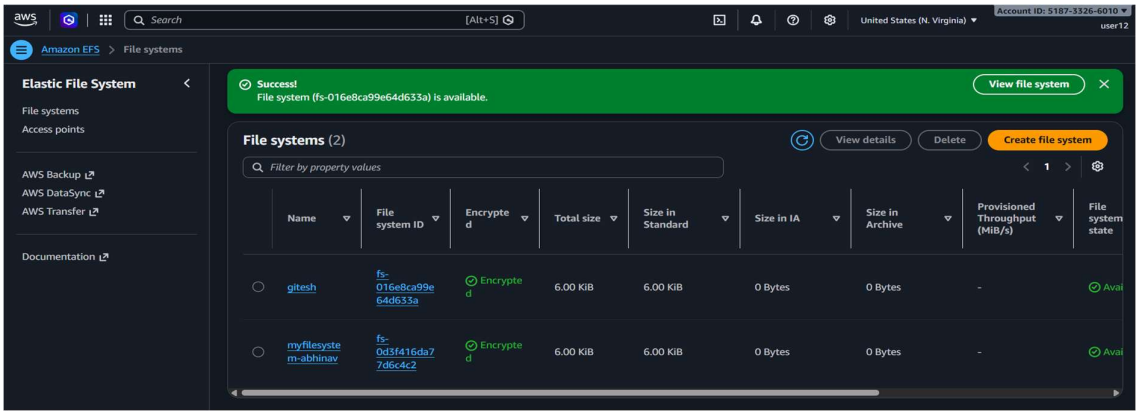
17. In File system Policy leave as default and Click on Next.




18. Review and Click on Create.



19. File System is created. Observe the total size of file system.



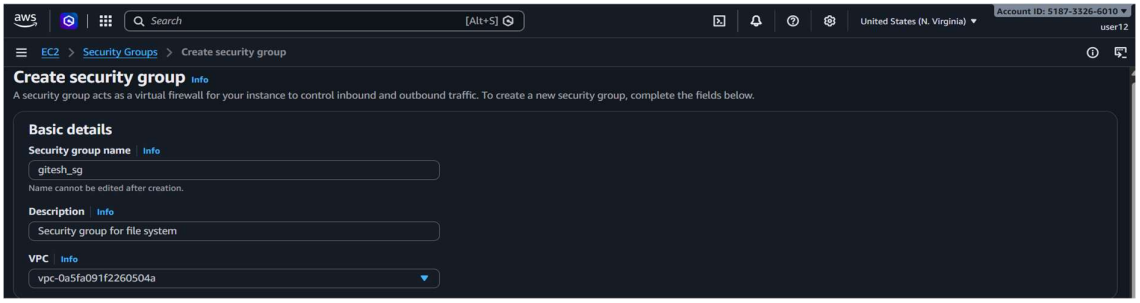

20. Open services menu -> Select EC2

21. Under network and security settings -> select Security Groups

22. Click on Create security group.

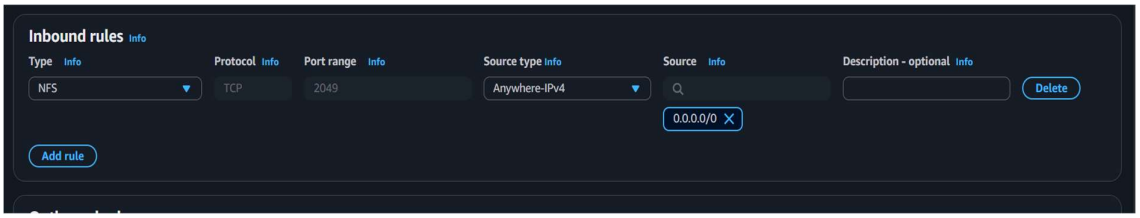
23. Create Security group wizard will open. Then enter the following details:

a) Security group name = Name of our choice
b) Description = Security group for file system
c) VPC = Give ID of the VPC in which file system is created.

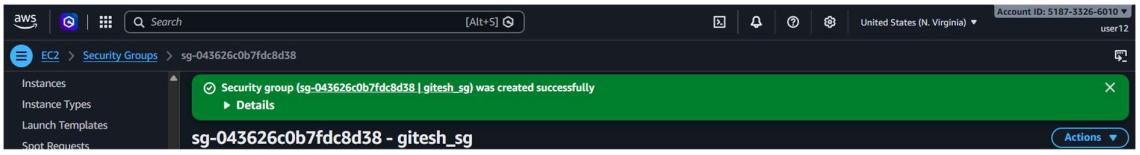



24. In Inbound rule click on Add rule.

25. Select Type = NFS and Source = IPV4 anywhere

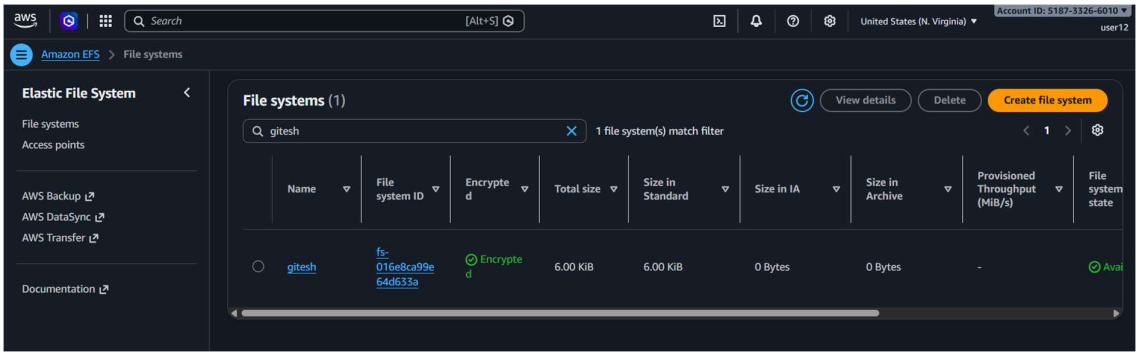



26. Leave rest as default and Click Create security group



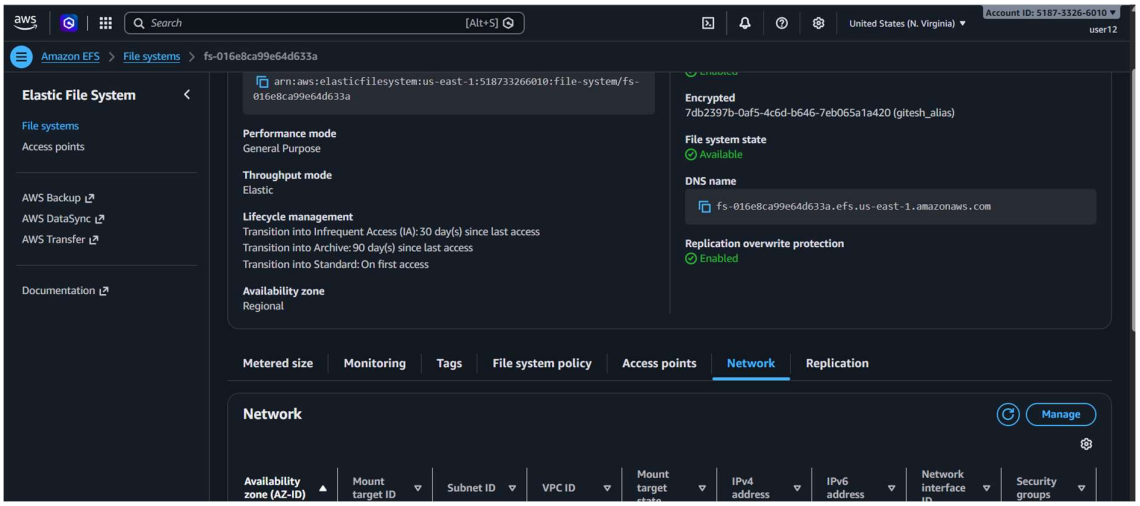
27. Open services menu -> Select EFS

28. Select your file system -> Click on View details

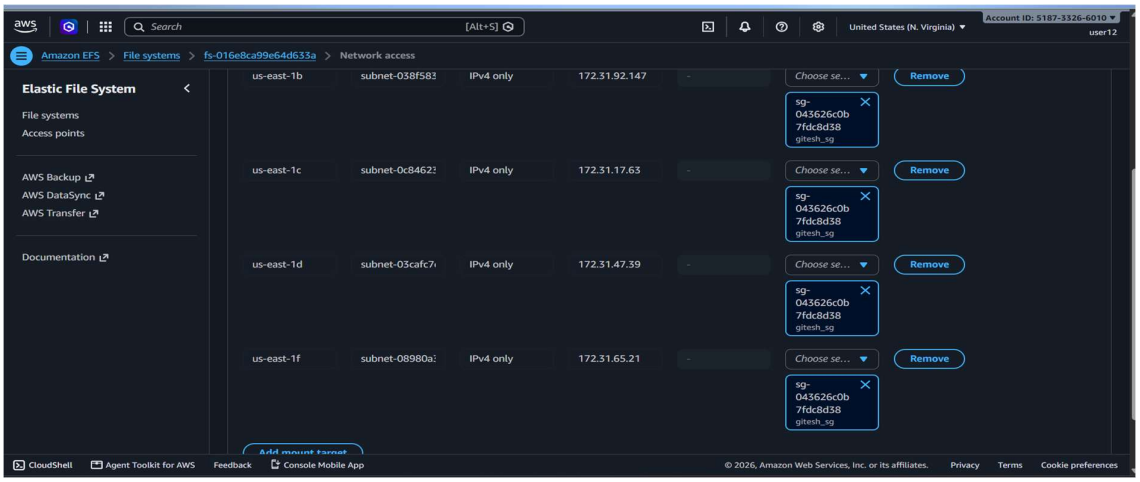



29. Click on Network Tab in details page

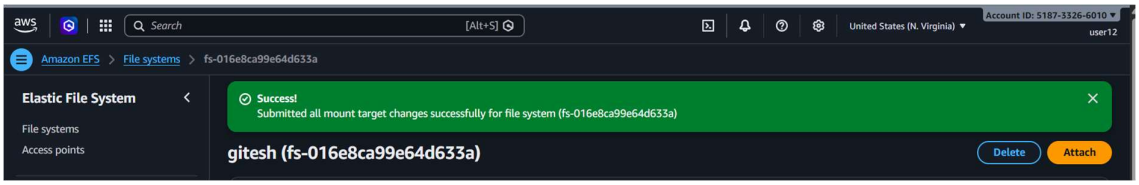
30. Under Network Tab -> Select Manage option.




31. In each Mount target -> remove default security group -> add security group created in previous steps with NFS protocol.



32. Click on Save.



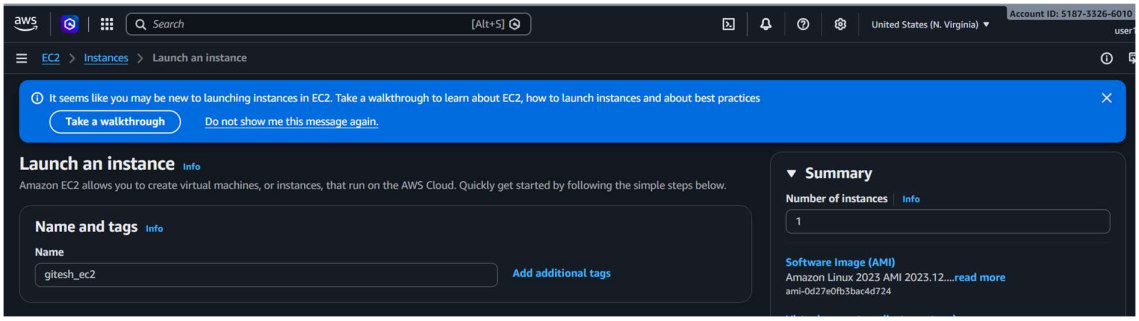

33. Open Service menu -> Select EC2.

34. EC2 dashboard will open -> Click on Instances.

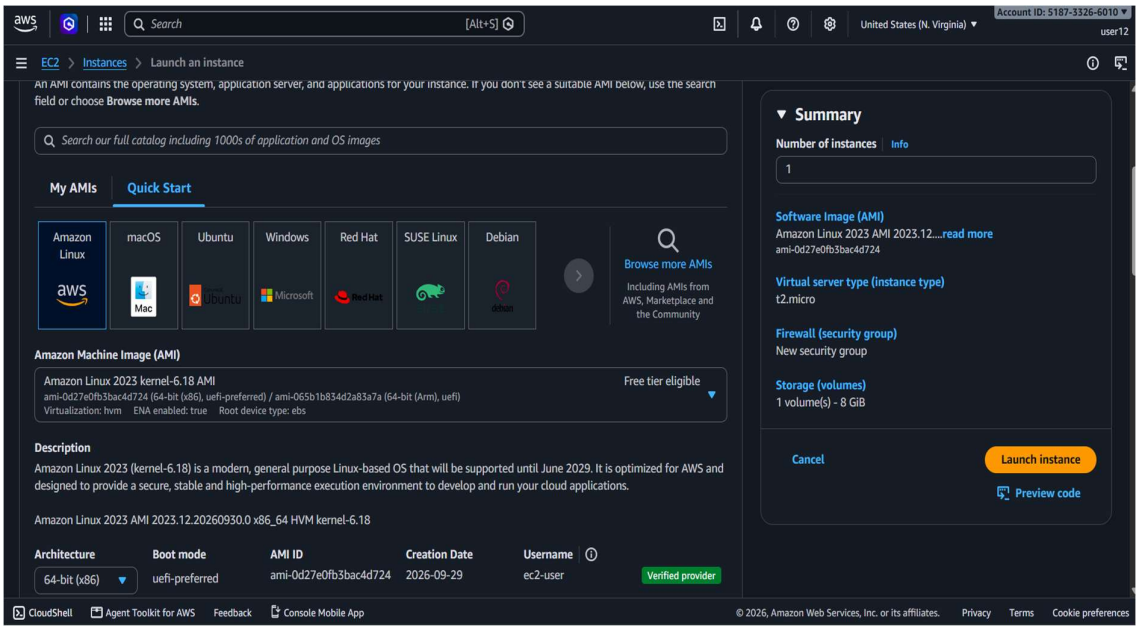
35. Then click on Launch Instances.

36. Launch wizard will open, then enter the following:

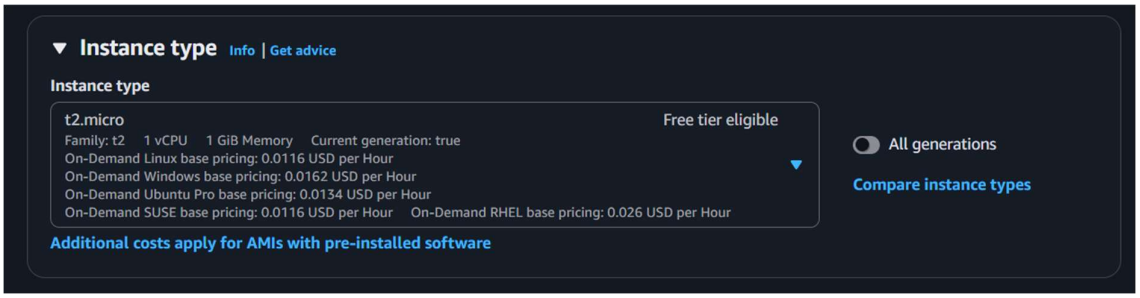
a) Name = FIRSTEC2
b) Number of instances = 1



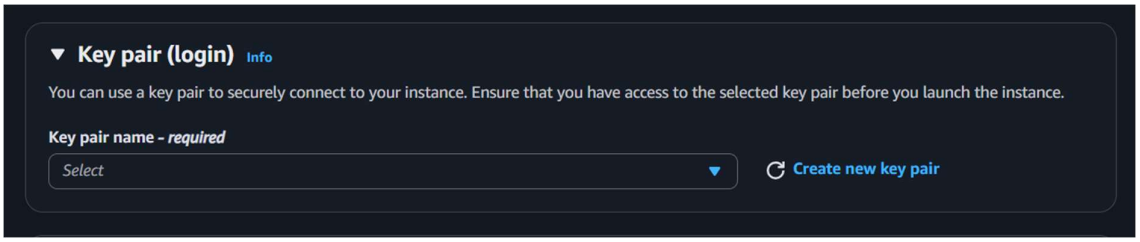

37. In application and OS image -> Amazon Linux



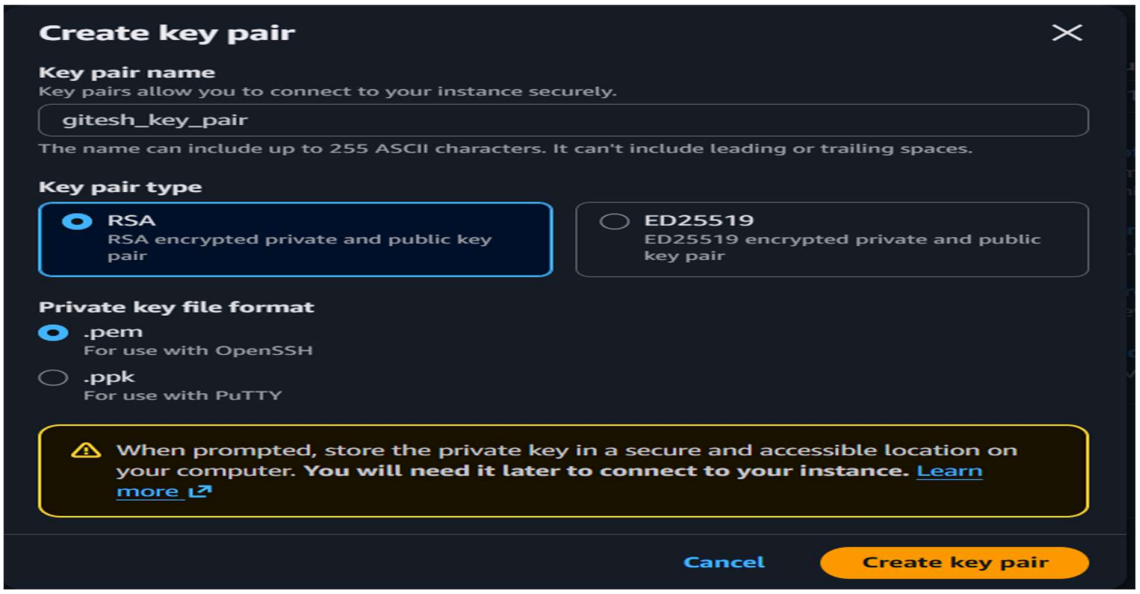
38. In Instance type -> Select t2.micro



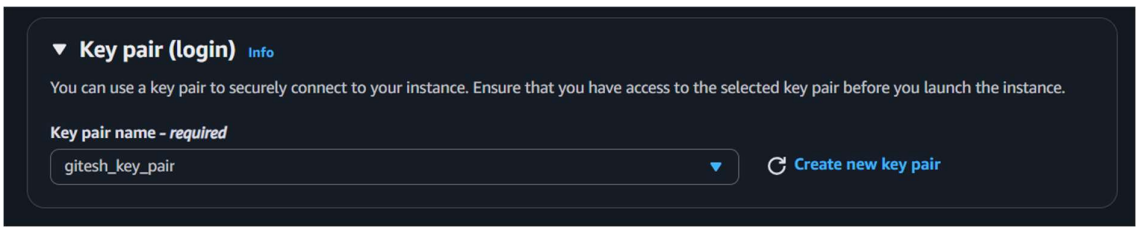

39. In Key pair -> click on Create new keypair



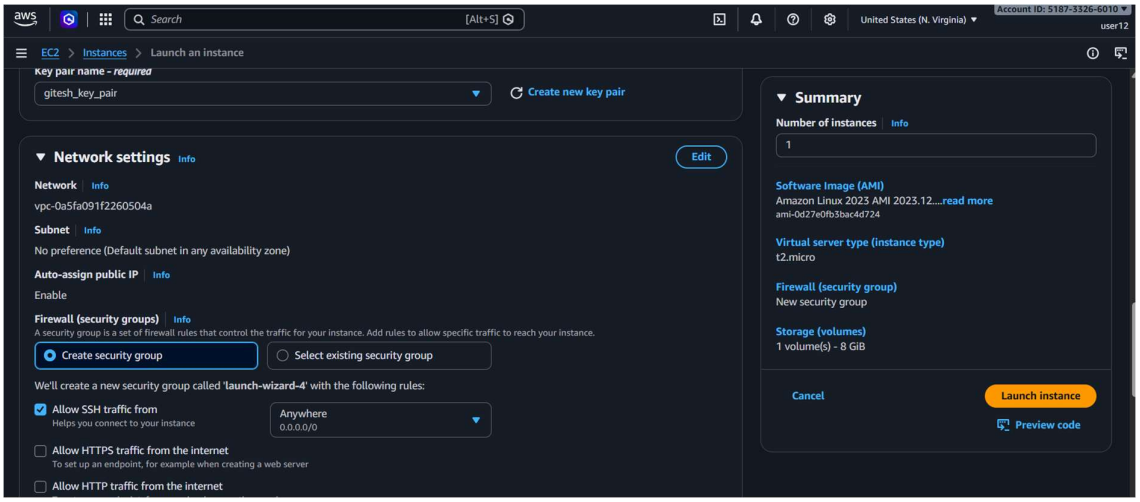

40. In Create key pair tab, Enter Name of your choice and click Create key pair




41. Choose the key pair selected in previous step.



42. In network settings -> Click on Edit.



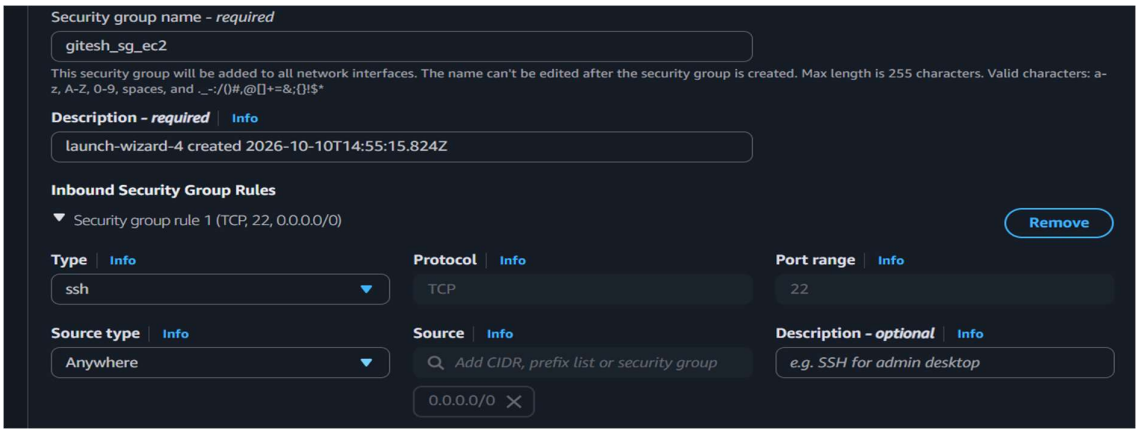

43. Select

a) VPC = VPC in which file system is created
b) Subnet = Choose subnet of your choice
c) Auto assign Public IP = Enable




44. Select Create Security group and Enter

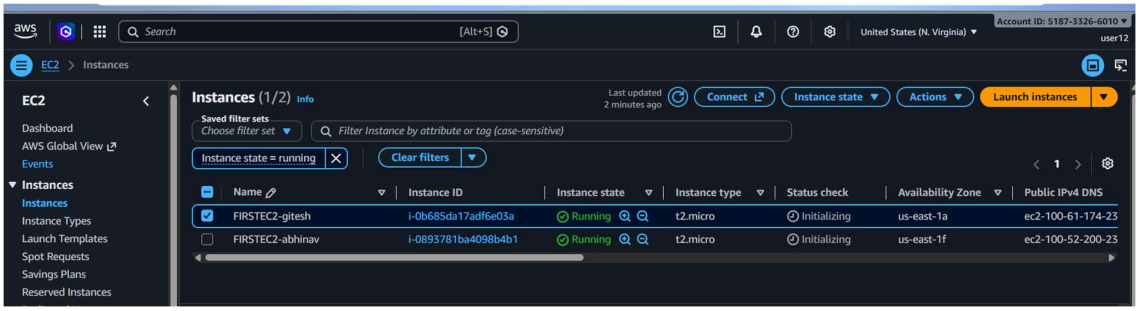
a) Security group name = Name of your choice
b) Type = SSH
c) Source = IPV4 anywhere



45. Leave rest as default -> Click Launch Instance

46. Click on View all instances.

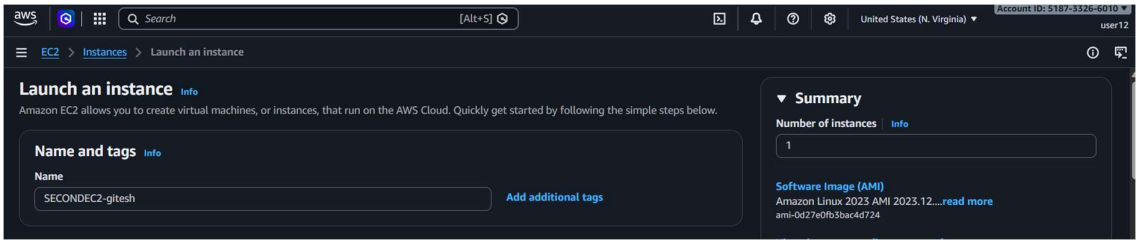
47. Notice that EC2 is launched successfully.

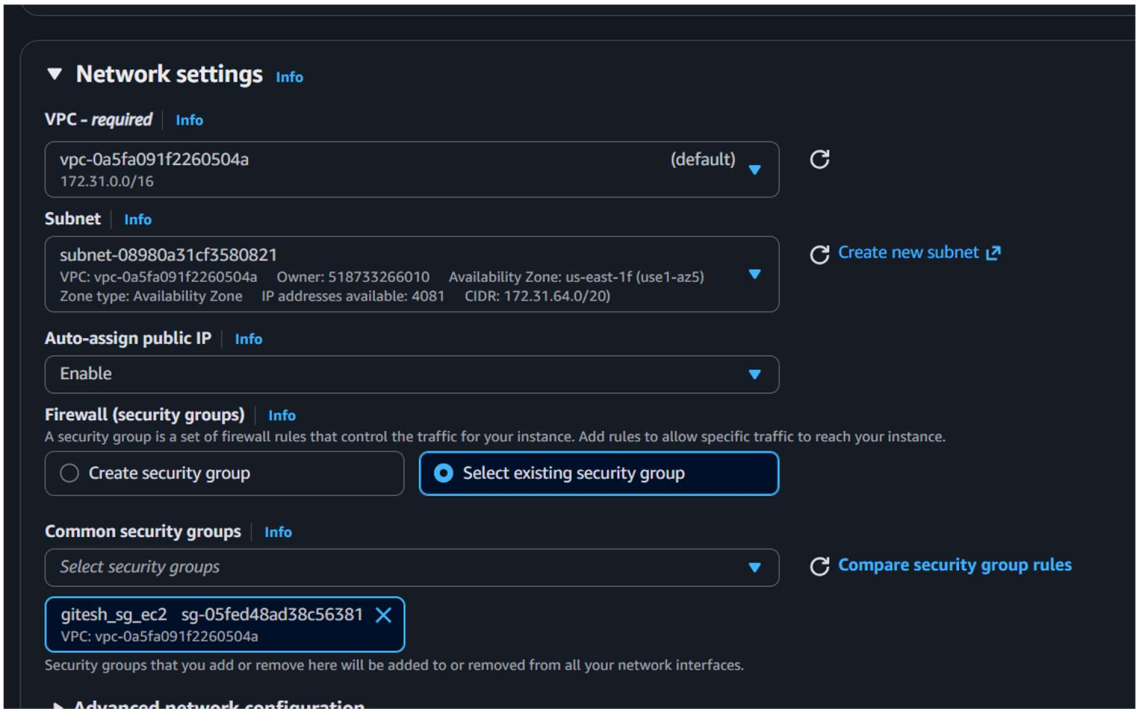



48. Launch one more EC2 repeat step number 36 – 46, In this EC2 give

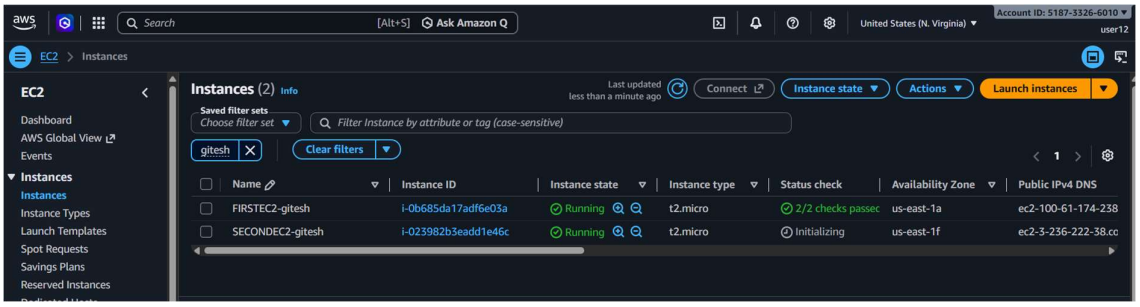
a) Name = SECONDEC2

b) Subnet = choose subnet in availability zone different from firstEC2

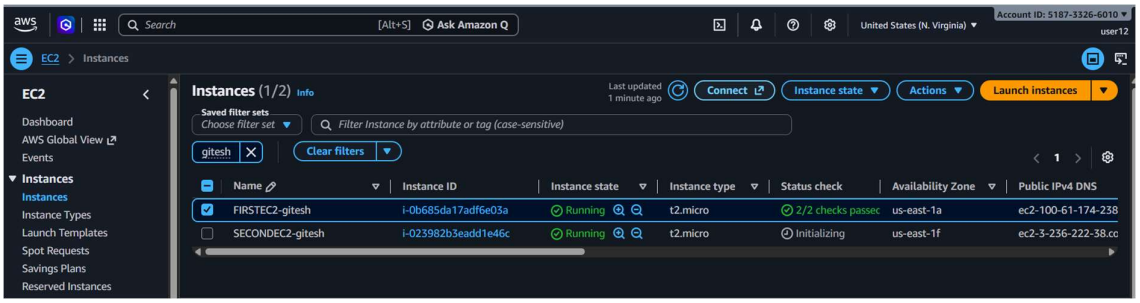


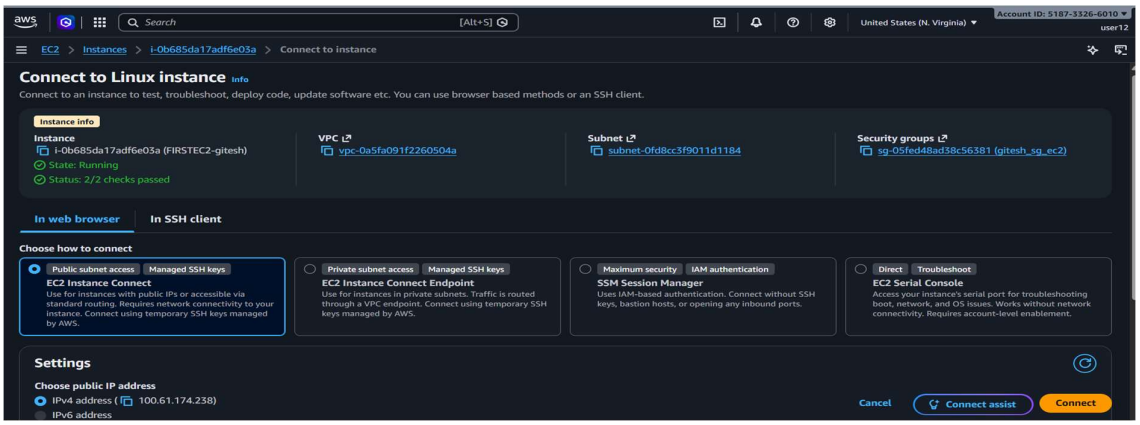
49. Observe both EC2 instances are launched in different availability zones.




50. Select FIRSTEC2 instance -> click CONNECT.




51. Select EC2 Instance connect -> click Connect

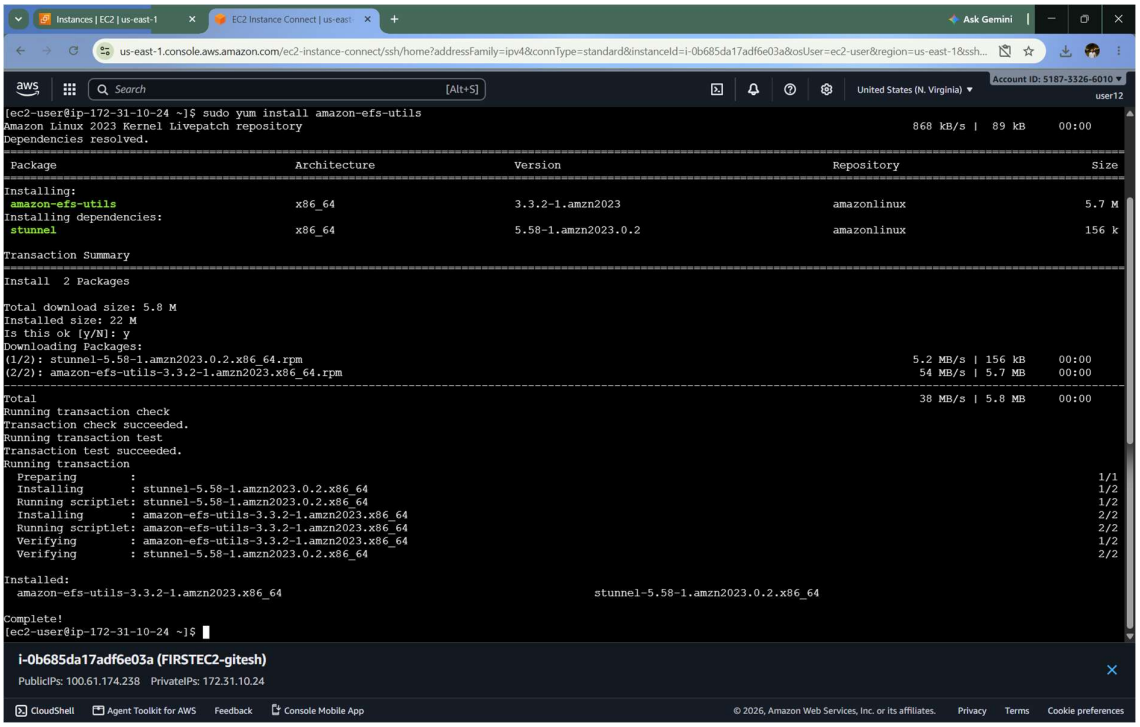



52. You are now inside FIRSTEC2

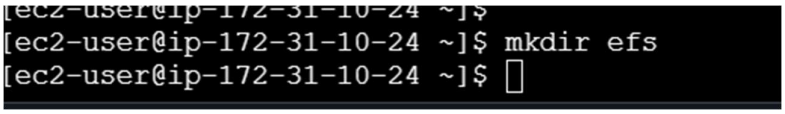


53. Enter the following commands

$ sudo yum install amazon-efs-utils

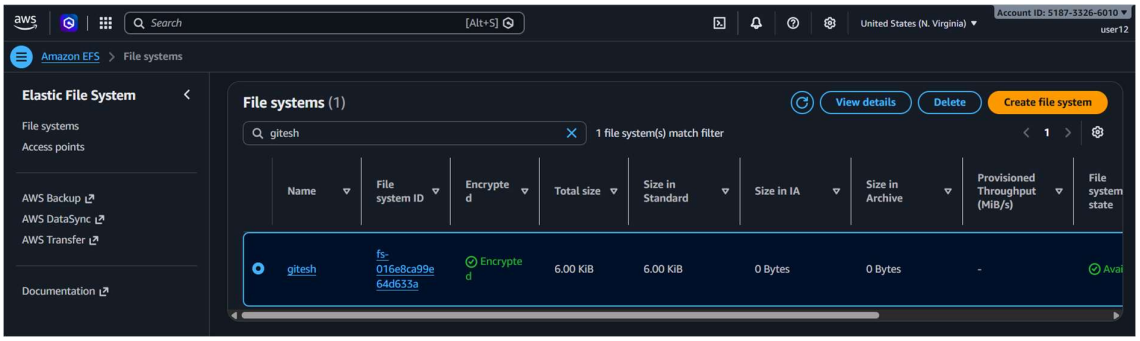



$ mkdir efs

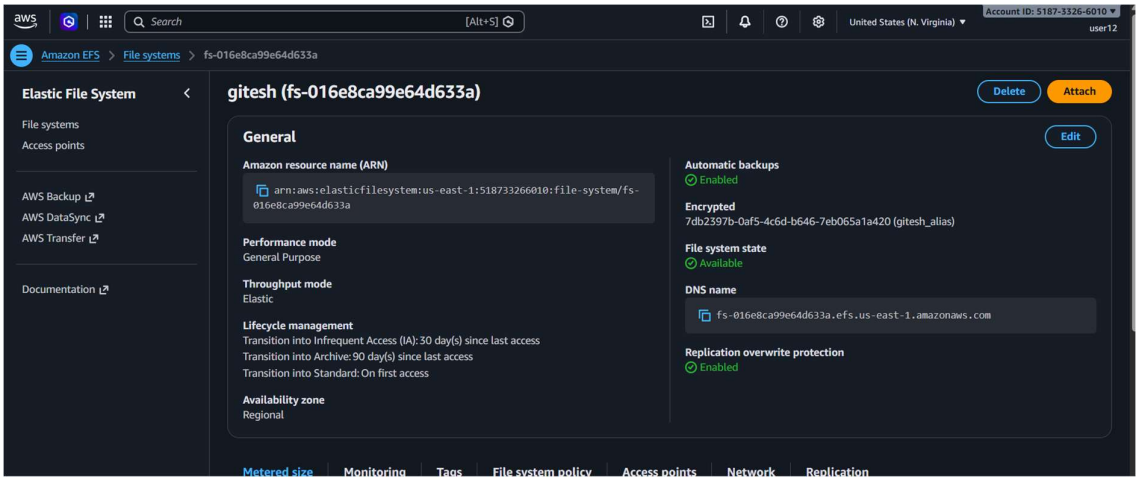



54. From services menu -> open EFS in new tab

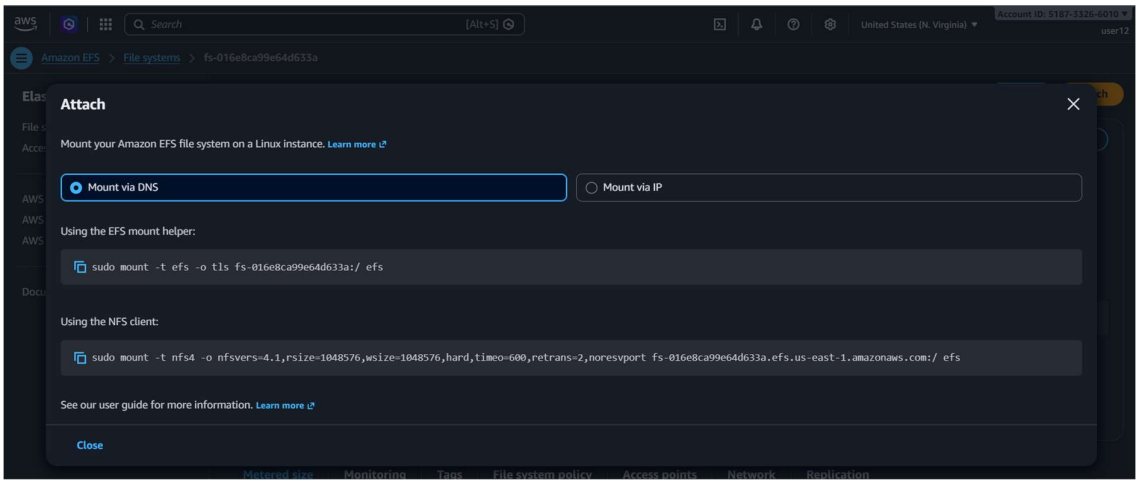
55. Select your EFS filesystem -> Click on View details



56. Click on Attach




57. Select Mount via DNS -> copy first command (This command has option parameter as tls which will enable encryption in transit).




58. Paste the command in EC2 CLI and press enter.

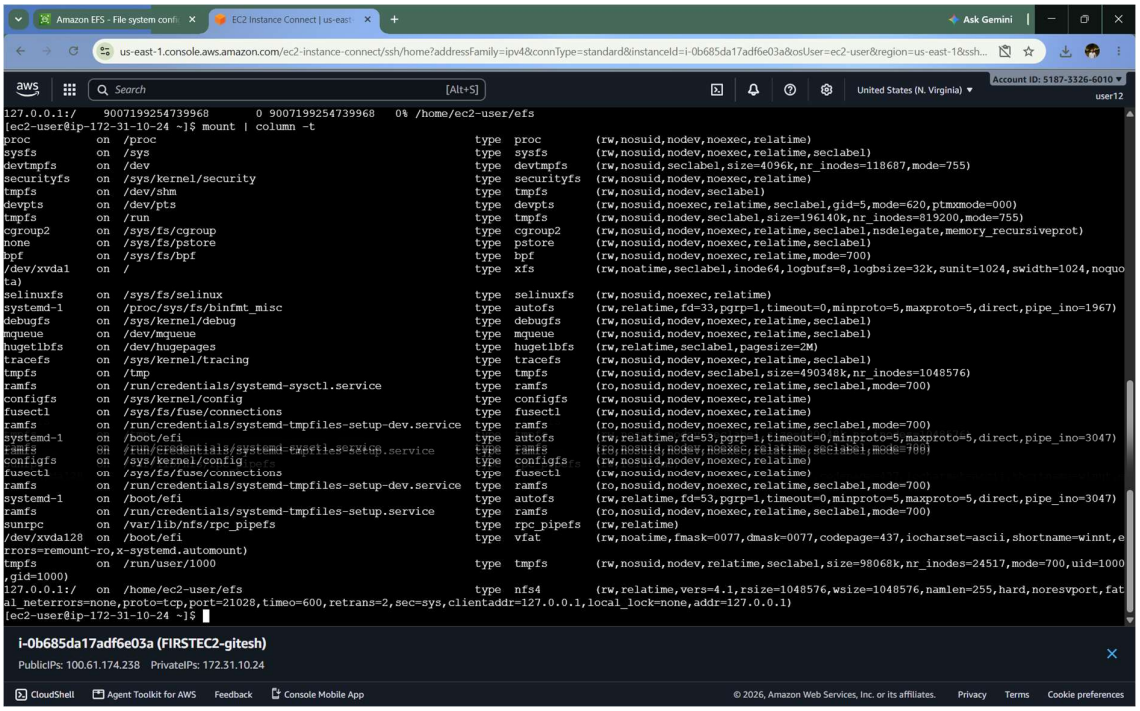
```
[ec2-user@ip-172-31-10-24 ~]$ mkdir efs
[ec2-user@ip-172-31-10-24 ~]$ sudo mount -t efs -o tls fs-016e8ca99e64d633a:/ efs
[ec2-user@ip-172-31-10-24 ~]$
```

59. Enter command $ df-k (And notice filesystem is mounted on /home/EC2-user/efs directory)

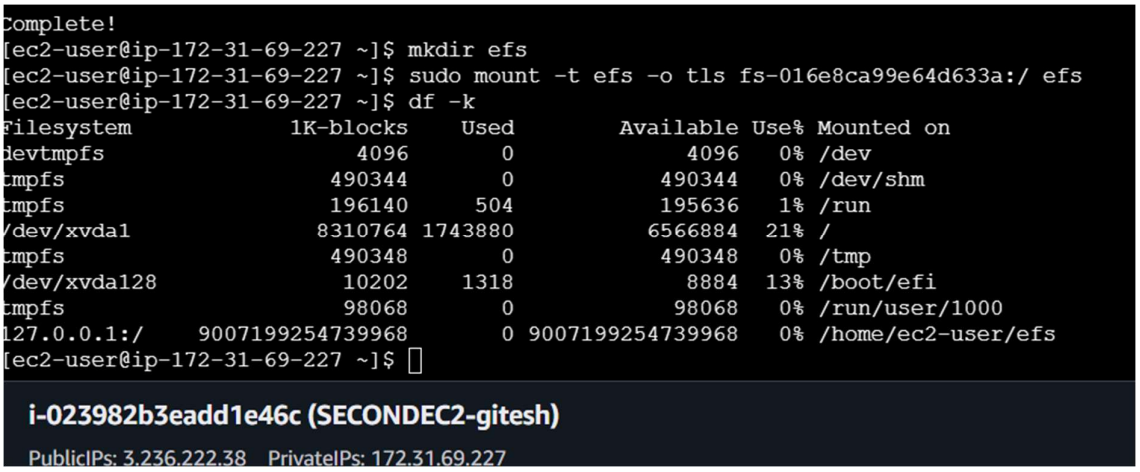
```
[ec2-user@ip-172-31-10-24 ~]$ df -k
Filesystem              1K-blocks    Used       Available Use% Mounted on
devtmpfs                     4096       0            4096   0% /dev
tmpfs                      490344       0          490344   0% /dev/shm
tmpfs                      196140     504          195636   1% /run
/dev/xvda1                8310764 1740300         6570464  21% /
tmpfs                      490348       0          490348   0% /tmp
/dev/xvda128                10202    1318            8884  13% /boot/efi
tmpfs                       98068       0           98068   0% /run/user/1000
127.0.0.1:/    9007199254739968       0 9007199254739968   0% /home/ec2-user/efs
[ec2-user@ip-172-31-10-24 ~]$
```

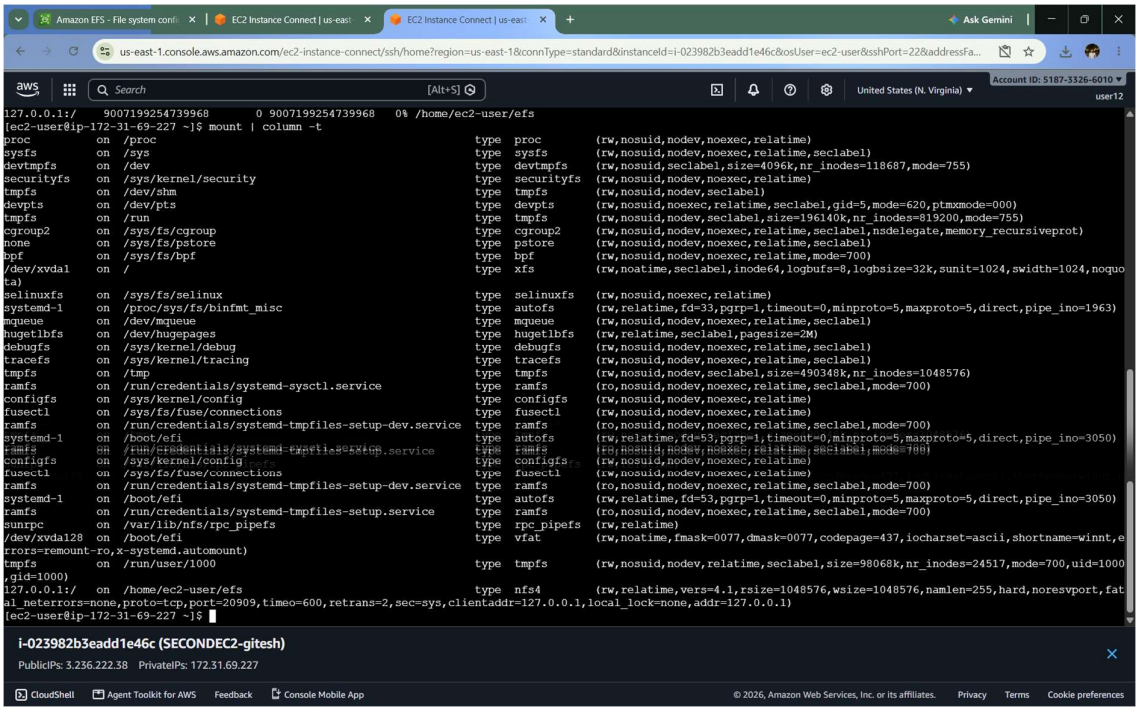
i-0b685da17adf6e03a (FIRSTEC2-gitesh)

60. Check if encryption at transit is enabled or not by giving command [$mount | column -t]




61. Now connect to SECONDEC2 and mount filesystem on it. REPEAT step number 51 to 59. And notice that file system is mounted on SECONDEC2.






62. Now in FIRSTEC2 CLI -> Enter command

$ cd efs (to go inside efs directory)




63. Enter command [ $ sudo touch file1 ( to create empty file named file1)]




64. Now in SECONDEC2 CLI -> Enter command [$ cd efs]



65. Enter command

$ ls (to list contents of the directory)




**You can access file1 created by FIRSTEC2 through SECONDEC2. Hence, shared file system is working with both EC2.**

66. In EC2 dashboard select both instances -> Instance state -> Terminate Instance

67. In EFS dashboard -> Select your file system -> click Delete